

OMER KOC

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Education

Northeastern University

M.Sc. in Data Science, 1st in Class, GPA: 3.96/4.00

Jan. 2023 – May. 2025

Boston, MA

Istanbul Technical University

B.Sc. in Industrial Engineering, 3rd in Class, GPA: 3.73/4.00

Sep. 2016 – Jun. 2021

Istanbul, Turkey

Experience

MATREU

Sep. 2024 – Dec. 2024

Data Analyst | *Amazon Ads, A/B Testing, Predictive Analytics, ETL Automation, Dashboarding*

Austin, TX

- Analyzed 20+ Amazon ad campaigns using Helium10 to measure ROI, engagement metrics, and product performance.
- Ran A/B tests on product listings and ad creatives, identifying high-performing segments to increase conversions.
- Built dashboards and trained ad models using 12 months of click and conversion data to track key KPIs.
- Streamlined ETL workflows to automate collection, cleaning, and integration of marketing and sales data.
- Developed predictive models to forecast sales trends, inform marketing strategies, and allocate ad spend more effectively.

Khoury College of Computer Sciences

Dec. 2023 – Jun. 2024

Graduate Teaching Assistant | *Data Modeling, Python, SQL, OLAP, MongoDB*

Boston, MA

- Guided 40+ students through data modeling, ER diagrams, database design, and normalization up to BCNF.
- Evaluated SQL queries, OLAP analysis, and semi-structured data tasks using Python, MongoDB, and XPath.
- Hosted weekly lab sessions on data cleaning, schema design, query tuning, and project debugging with Python and SQL.

Dynflux

Jan. 2022 – Dec. 2023

Data Science & Analytics Specialist | *Predictive Modeling, Feature Engineering, Python, SQL*

Boston, MA

- Developed predictive models using Python and SQL, optimizing marketing spend and increasing campaign ROI by 15%.
- Built Tableau dashboards to track funnel KPIs, customer engagement, and omnichannel sales trends for executives.
- Worked with developers to embed predictive insights in commerce apps, boosting user journeys and retention.
- Engineered features from raw clickstream and transaction data to improve model accuracy and segment targeting.

Turkcell

Dec. 2020 – Jan. 2022

Data Analyst | *Customer Segmentation, Ad Hoc Analysis, Power BI, Excel, SQL*

Istanbul, Turkey

- Monitored KPI trends for SME and Key Accounts, providing insights to support revenue targets and budget planning.
- Developed SQL scripts for customer segmentation and personalized campaigns, reducing churn and boosting ARPU.
- Consolidated multi-channel sales data and prepared executive dashboards and presentations using Excel and Power BI.
- Provided timely datasets and ad hoc analyses to McKinsey consultants to support company-specific strategic reports.
- Provided all departments with needed data, ensuring timely and accurate information flow throughout the organization.

Selected Projects

Urban Air Quality Forecasting | *Python, Time Series, XGBoost, LSTM, Tableau* 🔄

May. 2025

- Integrated 5 years of traffic and weather data for multiple cities to analyze urban air pollution and seasonal trends.
- Engineered features and built XGBoost and LSTM models to predict PM2.5, PM10, and NO2, reaching RMSE of 4.35.
- Deployed an interactive Tableau dashboard to show time series forecasts, feature impacts, and city pollution patterns.

PubMed XML to Relational & Analytical Database | *MySQL, ETL, SQL, R* 🔄

Feb. 2023

- Parsed PubMed XML and converted it to a normalized MySQL schema, ensuring data integrity and query efficiency.
- Implemented ETL pipelines using R to clean, load, and structure publication records for analysis.
- Wrote analytical SQL queries and custom R scripts to extract insights like top journals and quarterly article trends.

Intraday Electricity Market Price Forecast Using Machine Learning | *Python, Machine Learning* 🔄

Jun. 2021

- Analyzed 6 years of intraday energy market data, applying anomaly detection and correction with Facebook Prophet.
- Applied five feature selection methods for dimensionality reduction, identifying key predictors among 50+ variables.
- Achieved \$1.5 MAE with a CNN-LSTM hybrid, outperforming Bi-LSTM and GRU models in time-series forecasting.

Technical Skills

Languages & Tools: Python (Pandas, NumPy, Matplotlib, NLTK, Scikit-learn), SQL (advanced), MySQL, R, Tableau, Power BI, Excel, AWS (EC2, S3, Lambda), Git

Expertise: Data Cleaning & ETL, Relational Database Design, Statistical Analysis, Hypothesis Testing, A/B Testing, Time Series Forecasting, NLP, KPI Reporting, Dashboard Development